
Dwight Look College of ENGINEERING
TEXAS A&M UNIVERSITY

# Team 45: LES Parts Database UI
# Bi-Weekly Update 2

*Anthony Beneventi, Jamaal Bloom, Jaswin Jabbal*

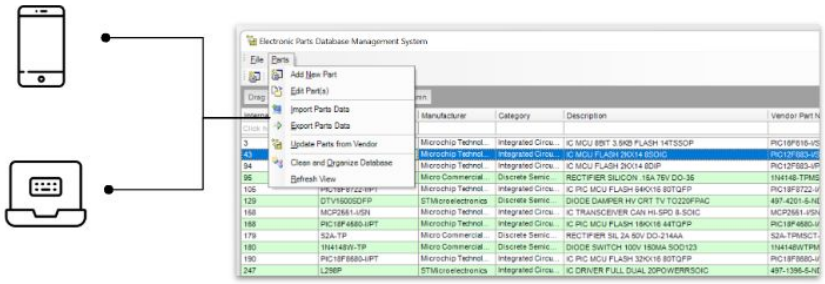
**Sponsor:** John Lusher

**TA:** Sabyasachi Gupta

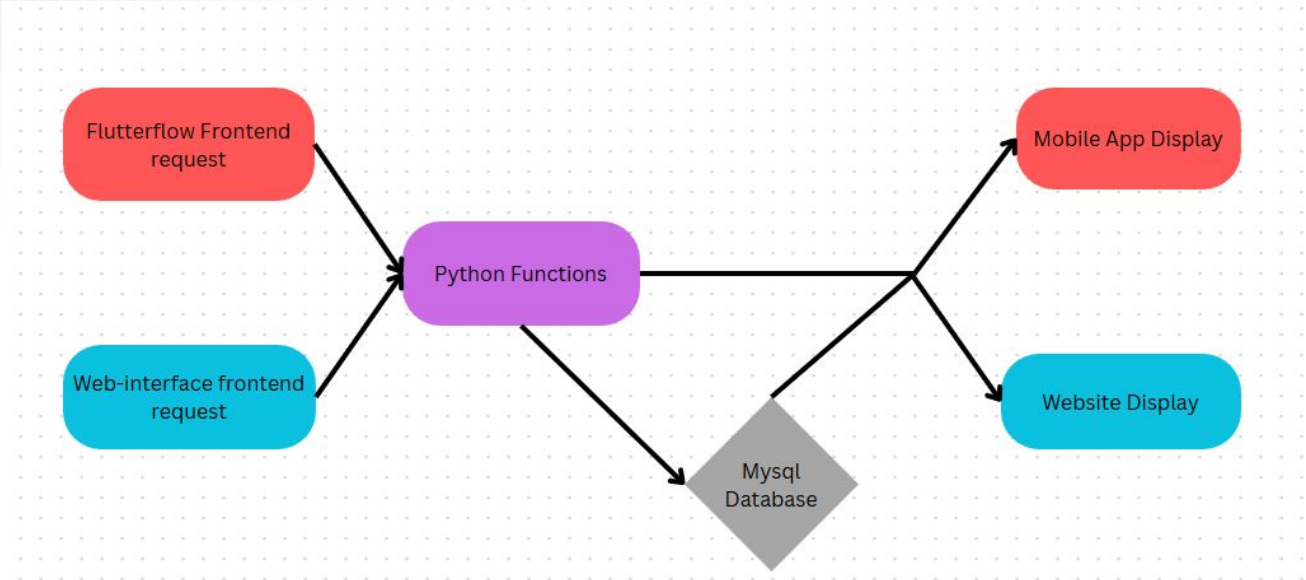
# Project Summary

- The current database requires manually updating parts and limits multiple users working at the same time.

- Develop a web and mobile application that will allow for the auto population and filtering of part information

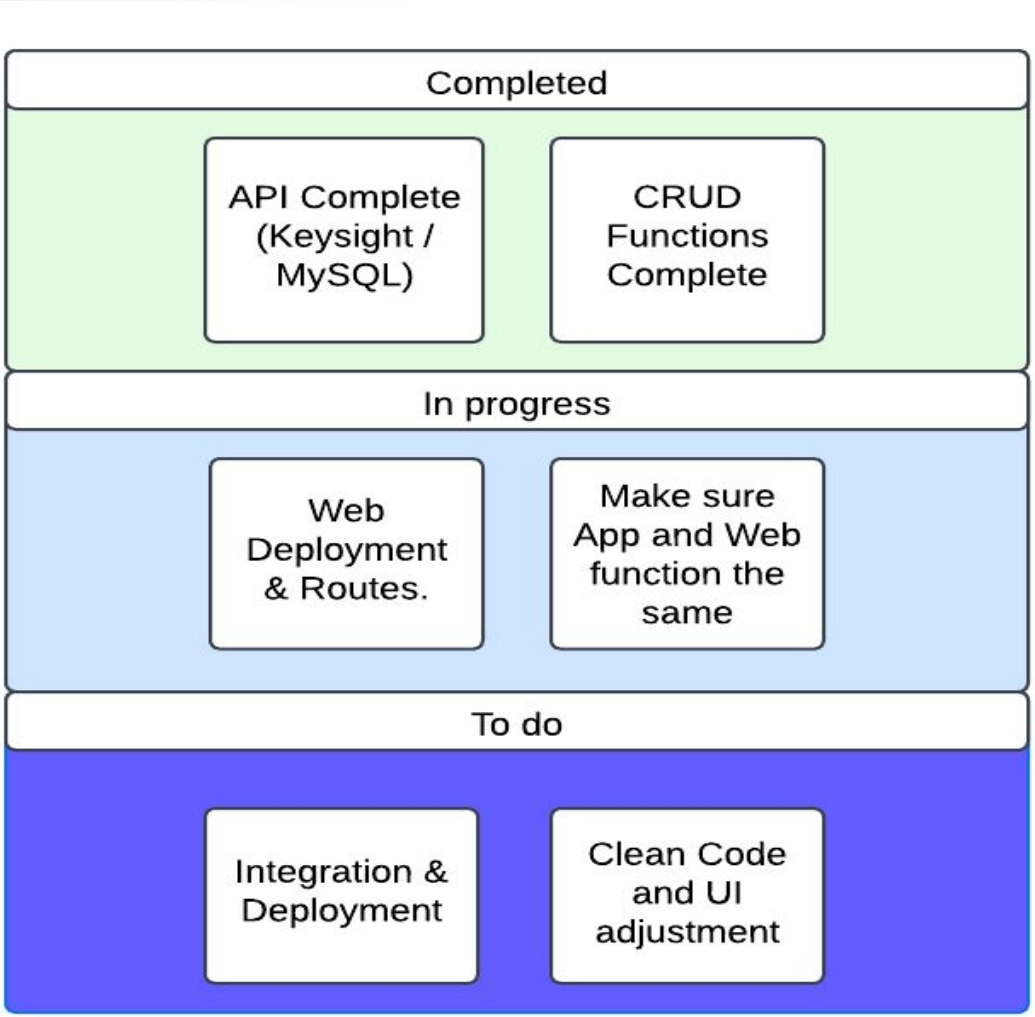


# Project/Subsystem Overview




Team:
Jamaal:
Anthony:
Jaswin:

# Project Timeline



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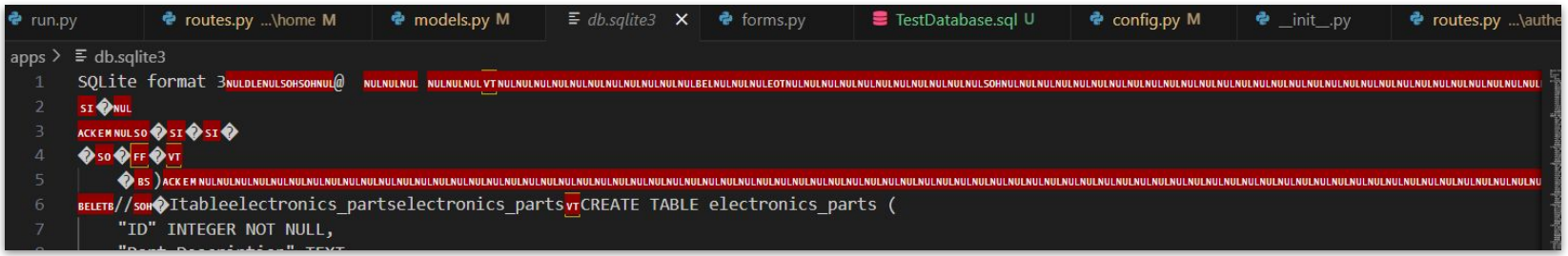
# Web Application UI

**Jaswin Jabbal**

| Accomplishments since Update 1<br>6+ hrs of effort | Ongoing progress/problems and plans until the next presentation |
|---|---|
| - Wrote new models based on MySQL schema<br>- Creating routes for CRUD operations - debugging | - **Route functionality/ validation**<br>- Frontend data consistency<br>- Deployment Environment |




*Exploring issuing of database creation within the app*

# Python Backend

**Jamaal Bloom**

| Accomplishments since Update 1<br>12 hrs of effort | Ongoing progress/problems and plans until the next presentation |
|---|---|
| • Converted code to callable API.<br>• Tested on Flutterflow | • Get and connect to an online database<br>• Format output for mobile and web use. |

Vercel

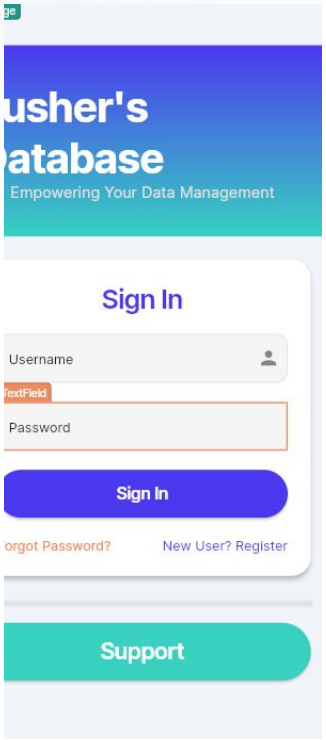
# IOS Application

**Anthony Beneventi**

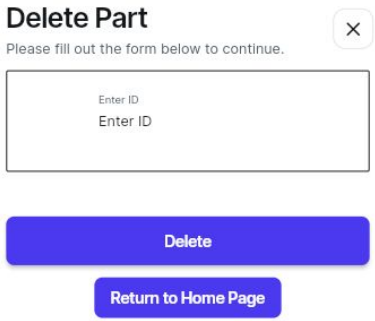
| Accomplishments since Update 1<br>8+ hrs of effort | Ongoing progress/problems and plans until the next presentation |
|---|---|
| • User Authority Feature<br>• Multiple User Functionality | • incorporate kesight API from WEB subsystem into IOS App.<br>• Integration into online database |

# IOS Application

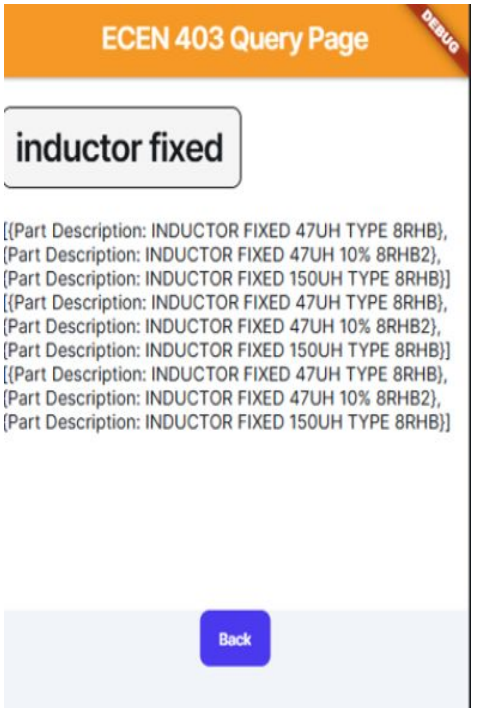
**Anthony Beneventi**




Functioning Forgot password + New User registration.




Delete function




Example of search query, “Inductor fixed.”

# Execution Plan

| Task Assignments | Owner | 1/13/25 | 1/20/25 | 1/27/25 | 2/3/25 | 2/10/25 | 2/17/25 | 2/24/25 | 3/3/25 | 3/10/25 | 3/17/25 | 3/24/25 | 3/31/25 | 4/7/25 | 4/14/25 | 4/21/25 | 4/28/25 | Due Date |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Finish Subsystems | Team | | | | | | | | | | | | | | | | | |
| Convert code to callable API | Jamaal | | | | | | | | | | | | | | | | | 1/27/25 |
| New models based on DB requirement | Jaswin | | | | | | | | | | | | | | | | | |
| Fix Update/Insert Butt | Anthony | | | | | | | | | | | | | | | | | |
| Connect with online database | Jamaal | | | | | | | | | | | | | | | | | 2/10/25 |
| Routes based on new models | Jaswin | | | | | | | | | | | | | | | | | |
| Integrate API | Anthony | | | | | | | | | | | | | | | | | |
| Format frontend and backend requests | Jamaal | | | | | | | | | | | | | | | | | 2/24/25 |
| Data Consistency | Jaswin | | | | | | | | | | | | | | | | | |
| Consolidate App | Anthony | | | | | | | | | | | | | | | | | |
| Finish integration | Team | | | | | | | | | | | | | | | | | |
| Implement failsafes | Jamaal | | | | | | | | | | | | | | | | | 3/17/25 |
| Deployment | Jaswin | | | | | | | | | | | | | | | | | |
| System validation | Anthony | | | | | | | | | | | | | | | | | |
| Finialize/Polish code | Jamaal | | | | | | | | | | | | | | | | | 3/31/25 |
| Deployment | Jaswin | | | | | | | | | | | | | | | | | |
| Optimize code | Anthony | | | | | | | | | | | | | | | | | |
| Final Presentation | Team | | | | | | | | | | | | | | | | | 4/14/25 |
| Showcase/ Demo | Team | | | | | | | | | | | | | | | | | 4/24/25 |

| Key | |
|---|---|
| *Planned* | |
| *In Progress* | |
| *Complete* | |
| *Behind Schedule* | |

# Validation Plan

| Pararagph # | Test Name | Success Criteria | Methodology | Status | Responsible Engineer(s) |
|---|---|---|---|---|---|
| 3.2.1.4 | Database Sync | The web and mobile applications uses the same database and changes visible from both | Attempt changes from multiple devices and ensure consistency | Untested | Jamaal |
| 3.2.1.1 | API Correctness | The API pulls correct information from the supplier | Compare recieved information with suppliers website | TESTED | Jamaal |
| 3.2.6.2 | Request Correctness | The backend correctly recieves and handles the frontend requests | Send frontend request from both applications and examine results | Untested | Jamaal |
| 3.2.1.3 | Multiple Requests | Application can handle multiple requests at the same time | Send multiple requests | Untested | Jamaal |
| 3.2.2.1 | Model Validation | All DB models align correctly with the updated schema used by Jamaal | Compare Flask SQLAlchemy models against the latest database schema | Untested | Jaswin |
| 3.2.4.1 | Route Functionality | Form submission routes to the correct display without errors | Manually fill out and submir forms in the application, verifying data correctness | Untested | Jaswin |
| 3.2.4.2 | Frontend Data Consistency | Submitted data is correctly displayed on a corresponding UI template | Manual validation by cross-checking displayed records with DB entries | Untested | Jaswin |
| 3.2.6.1 | Deployment Environment | All required dependencies are correctly installed and configured on the end machine | Validate with Docker, verify installation logs, ensure WGSI server | Untested | |
| 3.2.1.2 | CRUD Operation Test | Pulling apart information from the MySQL Database | Access a local MySQL Server with dummy datta | Untested | Anthony |
| 3.2.1.4 | Updated Database | The API Correctly updates the entire database | Checking previous and updated dababases to ensure completion | Untested | Anthony |
| 3.2.1.5 | Request Correctness | Having all requested components, such as User authority | Giving specific permission to individual User admins | Untested | Anthony |
| 3.2.1.3 | Application Deployed | Uploaded on the cloud and ready for use | Able to be sent for beta testing | Untested | Anthony |

Team 45: Anthony Beneventi, Jamaal Bloom, and Jaswin Jabbal

# Questions?